requires a knowledge of the signal and noise spectra *beforehand*, so that the filter's frequency response can be determined. This nonlinear technique uses the same idea, except that the Wiener filter's frequency response is recalculated for each segment, based on the spectrum *of that segment*. In other words, the filter's frequency response changes from segment-to-segment, as determined by the characteristics of the signal itself.

One of the difficulties in implementing this (and other) nonlinear techniques is that the overlap-add method for filtering long signals is not valid. Since the frequency response changes, the time domain waveform of each segment will no longer align with the neighboring segments. This can be overcome by remembering that audio information is encoded in frequency patterns that change over time, and not in the shape of the time domain waveform. A typical approach is to divide the original time domain signal into *overlapping* segments. After processing, a smooth window is applied to each of the overlapping segments before they are recombined. This provides a smooth transition of the frequency spectrum from one segment to the next.

The second nonlinear technique is called **homomorphic** signal processing. This term literally means: *the same structure*. Addition is not the only way that noise and interference can be combined with a signal of interest; multiplication and convolution are also common means of mixing signals together. If signals are combined in a nonlinear way (i.e., anything other than addition), they cannot be separated by linear filtering. Homomorphic techniques attempt to separate signals combined in a nonlinear way by making the problem *become* linear. That is, the problem is converted to the *same structure* as a linear system.

For example, consider an audio signal transmitted via an AM radio wave. As atmospheric conditions change, the received amplitude of the signal increases and decreases, resulting in the loudness of the received audio signal slowly changing over time. This can be modeled as the audio signal, represented by $a[\ ]$, being *multiplied* by a slowly varying signal, $g[\ ]$, that represents the changing gain. This problem is usually handled in an electronic circuit called an *automatic gain control* (AGC), but it can also be corrected with nonlinear DSP.

As shown in Fig. 22-11, the input signal, $a[\ ] \times g[\ ]$, is passed through the logarithm function. From the identity, $\log(x+y) = \log x + \log y$, this results in two signals that are combined by addition, i.e., $\log a[\ ] + \log g[\ ]$. In other words, the *logarithm* is the homomorphic transform that turns the nonlinear problem of *multiplication* into the linear problem of *addition*.

Next, the added signals are separated by a conventional linear filter, that is, some frequencies are passed, while others are rejected. For the AGC, the gain signal, $g[\ ]$, will be composed of very low frequencies, far below the 200 hertz to 3.2 kHz band of the voice signal. The logarithm of these signals will have more complicated spectra, but the idea is the same: a high-pass filter is used to eliminate the varying gain component from the signal.